InstaRoad: Detecting Road Networks from Medium Resolution Satellite Imagery - Literature Review

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Abstract

We outlined the difficulty of obtaining ultra-high-resolution remote sensing images to train highly accurate road segmentation models. We explored alternative approaches that leverage the freely available Sentinel-1 and Sentinel-2 images. We analysed various data preprocessing techniques to improve segmentation performance by enhancing data quality. Finally, we compared the exisiting methodologies and concluded that fine-tuning a geospatial-foundational model is the most tentative approach.

1 Introduction

Accurate road network extraction from remote sensing imagery has applications in urban and rural development research, infrastructure planning, ecological studies, and disaster response. However, much of the existing work in this area relies on high-resolution satellite imagery and proprietary datasets that are either incomplete in geographic coverage or restricted behind paywalls [3].

Sentinel-1 and Sentinel-2, operated under the EU's Copernicus Programme, offer free access to satellite images with medium spatial resolution and high temporal resolution [9]. Leveraging these freely available data sources could democratise access to road extraction models, removing a key barrier for researchers and practitioners who lack the budget for commercial imagery.

Traditionally, road extraction from remote sensing images has been performed using statistical learning techniques such as random forests, support vector machines, and decision trees [3]. More recently, the research focus has shifted towards deep neural networks for automatic feature extraction, with architectures such as CNNs, Vision Transformers, Swin Transformers, and Mamba serving as backbones [3].

Despite these advances, limited work has addressed road extraction from freely available satellite images. This gap stems from two linked challenges. First, at Sentinel-2's 10m spatial resolution, non-major roads are often only one pixel wide, making standard segmentation difficult [21]. Existing literature tends to focus on applying segmentation models to ultra-high-resolution imagery (sub-1m), and the methods developed in that setting do not transfer directly to medium-resolution inputs [15].

Second, because few studies target this resolution range, there is a lack of established, labeled datasets suited to road extraction from Sentinel imagery. While crowd-sourced sources such as OpenStreetMap can provide road labels, these are often incomplete or noisy and require additional curation [1, 10]. Several data preprocessing and labelling strategies have been proposed to address these issues individually, including multi-modal fusion of SAR and optical data [14], band fusion to produce uniform 10m multi-spectral inputs [24], and super-resolution approaches that train models to predict high-resolution masks from medium-resolution inputs [1, 21]. Identifying, devising and testing techniques to enhance data quality is a core challenge of this project.

To manage the end-to-end pipeline, we use InstaGeo, an open-source geospatial machine learning framework developed by InstaDeep that provides data preprocessing, model training, and inference infrastructure [26]. Its data processing module can stack Sentinel-1 and Sentinel-2 imagery into a multi-temporal format compatible with PrithVi [12], a foundational geospatial model, while its teacher-student training component and full-stack interface help streamline the workflow. Its highly modularised nature also eases modifications, allowing us to swap out or tweak modules as needed.

This literature review is structured to address these specific challenges. Section 2 examines the properties and limitations of the relevant satellite and vector datasets. Section 3 reviews existing literature on the model training pipeline, from data preprocessing, deep learning objectives and techniques, to evaluation. Finally, Section 4 and 5 discusses the findings to identify the most viable methodology for our specific road extraction task.

2 Dataset

2.1 Sentinel 1 and Sentinel 2

The European Space Agency's Copernicus Programme provides freely accessible imagery with high-revisit time via Sentinel-1 and Sentinel-2, presenting unique physical trade-offs for road extraction [9].

Sentinel-1 utilises Synthetic Aperture Radar (SAR), which is weather-agnostic and highly effective at differentiating surface types and identifying networks obscured by tree canopies. However, its susceptibility to terrain distortion makes it less viable for detecting roads in steep, mountainous regions and urban centres.[14, 25].

On the other hand, Sentinel-2 provides multi-spectral optical imagery, but its maximum spatial resolution of 10m poses a significant obstacle for identifying narrow road networks. Because rural and non-major roads are frequently narrower than a single pixel at this scale, raw Sentinel-2 images generally yield poor segmentation performance. Consequently, literature focusing on these datasets often incorporates advanced preprocessing, super-resolution, or multi-modal fusion techniques to produce a functional road extraction system [2, 14, 24].

2.2 OpenStreetMap

Crowd-sourced data such as OpenStreetMap (OSM) provides a promising avenue for creating large-scale labeled datasets. OSM provides a public API to query labeled road vector data for a given

area, which includes 2D road networks and other geo-information. By buffering the road network edges, these vectors can be rasterised to obtain 2D binary ground-truth masks for training image segmentation models, similar to the approach proposed by Ayala et al. [1].

However, due to sensing noise (such as cloud coverage and shadows) and labeling noise inherent to OSM (such as omission errors where roads are unmapped), their method requires human-curated validations to ensure data quality. To account for time and resource constraint, Ayala et al. proposed first pre-training the models on the massive, unfiltered OSM dataset, and then fine-tuning and testing the models on filtered, high quality data. [1]

2.3 Expert Curated Datasets

Standard benchmarking datasets for road extraction almost exclusively feature ultra-high-resolution imagery [4, 19], making them unsuitable for our medium-resolution constraints. While expert-curated datasets for Africa do exist [17], visual inspection via QGIS revealed significant spatial misalignment between the provided vector labels and the actual road surfaces in the satellite imagery, rendering them unviable for model training.

Karnik et al. curated an accurately labeled, manually filtered Sentinel-2 road network dataset for India based on the methodology described above. Because this dataset directly addresses alignment issues and explicitly includes rural road networks, it serves as a highly relevant, high-quality ground-truth source for our specific extraction task.

3 Comparative Review of Literature

3.1 Data Preprocessing

3.1.1 Fusion of Sentinel 1 SAR with Sentinel 2 Optical Images. Continuing with the foundational work by Ayala et al. [1], Lin et al. [14] addressed the modality gap by proposing the United U-Net (UU-Net), which fuses Sentinel-1's SAR (VH and VV) and Sentinel-2's optical (RGB and NIR) bands. Because the physical imaging mechanisms differ vastly, simply stacking the raw layers proved suboptimal. Instead, the authors normalised and concatenated the extracted features at a deeper encoder level. This multimodal fusion effectively leverages SAR's ability to detect differences in surface types and optical data's robustness against terrain distortion, significantly outperforming single-modality baselines (S1-U-Net and S2-U-Net). We can potentially include their findings as a preprocessing step in our training pipeline.

3.1.2 Fusion of Sentinel 2 High Resolution Bands with Low Resolution Bands. Most existing literature on road extraction from Sentinel 2 data, only focus on using the 10m RGB bands and NIR band due to differing spatial resolutions of other bands.

Wang et al. [24] show that we can fuse the six 20m bands of Sentinel-2 with the four 10m bands to produce a set of ten bands at with 10m spatial resolution. To achieve this, they implemented a geostatistical algorithm called Area-to-Point Regression Kriging (ATPRK) [23] to downscale the six 20m bands by referencing the four 10m bands [2].

The authors noted that the resulting 10-band-10m images contain greater spatial detail, allowing elongated features and urban fabric to be observed much more clearly [24]. Hence, their work provides another tentative way to enhance our data quality.

3.1.3 Filters to Enhance Features. Different frequency bands can enhance different features captured by the satellite. One such example is the NIR band, which is often included alongside the RGB bands in the exisiting literature [1, 3, 14, 15, 21]. This is because NIR bands can help differentiate impervious surfaces from others: for example, roads from rivers and bare soils [22]..

Furthermore, combinations of these bands can be formulated into spectral indices (filters) to further enhance object identification [6]. Since rural road extraction models frequently misclassify rivers and farmlands as road networks, integrating indices specifically designed to detect vegetation and moisture could serve as a computationally inexpensive mechanism to reduce false-positive rates prior to model training.

3.1.4 Super-resolution to Enhance Satellite Image Quality. Another way to enhance data quality is to directly upsample the images. With deep learning, this is normally done via Generative Adversarial Networks and Diffusion Models. However, the problem with these generative models is that they are prone to hallucination, which introduces critical false-positive artifacts into road network predictions.

Aybar et al. [2] proposed SEN2SR, a low hallucination super-resolution framework for Sentinel 2 images. Their framework leverages feeding synthetic high resolution and low resolution satellite image pairs to a traditional deep neural network. The light weight model uses a CNN [13] as the backbone, and the more complex one combines Mamba [5] and Swin Transformer [16].

Our informal, preliminary testing indicates that applying foundational models like the Segment Anything Model (SAM) [11] to these super-resolved outputs yields better identification than raw Sentinel-2 imagery. Therefore, we may incorporate their framework as another step in our preprocessing pipeline, if we can adequately address the additional computationa cost.

3.2 Deep Learning for Road Extraction on Satellite Images

3.2.1 U-Net and Multimodal Fusion. Deep learning approaches for road segmentation are frequently based on the U-Net architecture. As summarised in Table 1, Zhang et al. [27] pioneered the use of deep neural networks specifically for road segmentation on Sentinel-1 SAR imagery using a standard U-Net. However, the primary limitation of this early work is its opaque evaluation methodology, as the construction process for the evaluation dataset is not outlined and the scores are suspiciously high .

Subsequent research expanded on this baseline by incorporating multimodal data. Both Ayala et al. [1] and Lin et al. [14] demonstrated that U-Net models trained on fused Sentinel-1 SAR and Sentinel-2 optical data outperform single-modality baselines. Lin et al. achieved this by fusing the SAR and optical modalities at the encoder level to create the United U-Net (UU-Net), though their work lacks explicit ablation demonstrating the isolated effect of this fusion mechanism.

Ayala et al. utilised U-Net and ResNet-34 architectures to demonstrate that models can directly predict super-resolved masks from

these fused inputs, thereby improving mask quality. To achieve this, they created a dataset from rasterised OpenStreetMap (OSM) masks filtered by human experts; however, the unfiltered OSM ground truth inherently introduces labelling noise and omission errors.

3.2.2 Knowledge Distillation. To directly address the limited spatial resolution of Sentinel imagery, Sirko et al. [21] implemented a teacher-student paradigm using the HRNet architecture. Their approach successfully distilled labels from a high-resolution teacher model to a low-resolution student model. While this mitigates the resolution constraint during inference, the methodology strictly requires access to parallel, corresponding high-resolution imagery during the training phase, which limits its applicability under a data-scarce context.

Parallel datasets for Sentinel-2 and high resolution satellite images have been curated specifically for this. SEN2VENμS with spatially aligned Sentinel-2 image and its corresponding VENμS image is such an example [18].

3.2.3 Foundational Models. Training specialised deep neural networks from scratch requires computational resources that are often unavailable for standard research projects. To bypass this limitation, recent studies explore adapting large, pre-trained foundational models. Hetang et al. [7] paired a pre-trained Segment Anything Model (SAM) [11] encoder with geometry and topology decoders, achieving state-of-the-art road graph extraction. However, this approach is highly dependent on ultra-high-resolution datasets (SpaceNet [4] and CityScale [**?**]), and its applicability to 10m Sentinel data remains untested.

Given our specific hardware and data constraints, adapting a geospatial foundational model natively pre-trained on medium-resolution satellite imagery offers a practical alternative to training models from scratch. For example, TerraMind [8] is a recently introduced multimodal foundational model pre-trained directly on aligned Sentinel-1 and Sentinel-2 data. Leveraging a pre-trained model of this nature allows us to utilise established feature representations for downstream fine-tuning, serving as a tentative and resource-efficient solution for our road extraction pipeline.

3.3 Evaluation

To measure the quality of surface segmentation, the literature predominantly relies on standard pixel-wise statistical metrics, including Accuracy, F1-Score, Precision, Recall, and Intersection over Union (IoU/mIoU). Conversely, to evaluate the topological continuity of road centre line extraction, specialised connectivity metrics such as Average Path Length Similarity (APLS) and Customised Connectivity (CC) are required. A comprehensive review of these evaluation formulations is provided by Liu et al. [15].

4 Discussion

The literature demonstrates a fundamental incompatibility between standard segmentation architectures and medium-resolution satellite imagery for narrow road extraction, particularly in rural contexts. Because these narrow road features are visually indistinct at a 10m spatial resolution, applying vanilla models like U-Net directly to raw Sentinel data yields uniformly poor performance [1, 14]. Consequently, physical feature enhancement and multi-modal synthesis are not merely optional performance boosts; they are mandatory prerequisites for our task.

To bridge this resolution gap, this review identified highly complementary preprocessing strategies. First, fusing Sentinel-1 SAR with Sentinel-2 optical data is critical for our specific objective of rural road extraction. SAR inherently detects unpaved and canopy-obscured networks, while optical data counteracts SAR's vulnerability to terrain distortion [1, 14, 25]. Second, applying Area-to-Point Regression Kriging (ATPRK) to fuse Sentinel-2's 20m and 10m bands explicitly enhances elongated urban features [24]. Because ATPRK is a deterministic, parallelisable algorithm, it can be computed efficiently via High-Performance Computing (HPC) [20], effectively enhancing spatial detail without adding too much overhead to the deep learning training phase.

Furthermore, while the literature diverges on whether super-resolution is most effective as an explicit preprocessing step [1] or an implicit network objective [21], our time and resource constraints dictate our methodology. Training complex, implicit super-resolution models from scratch requires ultra-high-resolution data. Therefore, employing an established, low-hallucination framework like SEN2SR [2] to directly upsample Sentinel-2 images prior to model training is the most pragmatic approach.

This computational constraint also dictates our architectural selection. Although U-Net is the typical baseline (Table 1), training these networks from scratch on massive datasets is unfeasible. We must instead leverage geospatial foundational models via fine-tuning. However, model utility is strictly bounded by pre-training data. For instance, PrithVi [12], the default model within the InstaGeo framework [26], is pre-trained on 30m HLS imagery, rendering it too coarse to identify 10m road geometries. Similarly, generalist models like SAM [11] lack native multi-spectral understanding.

Conversely, TerraMind offers a highly viable alternative because it is natively pre-trained on aligned Sentinel-1 and Sentinel-2 data. Based on these deductions, the most promising methodology for our project is to fine-tune a customised decoder attached to the TerraMind encoder. By feeding this model preprocessed (fused and super-resolved) Sentinel data, and evaluating it against the manually filtered ground truth masks curated by Karnik et al. [10], we can possibly achieve robust road extraction within our strict resource limits.

5 Conclusion

This review addressed the core challenge of extracting narrow road networks from free, medium-resolution (10m) satellite imagery. The literature indicates that raw Sentinel data lacks the spatial detail required for standard segmentation models to succeed. To overcome this, our findings suggest a hybrid methodology: first, applying deterministic preprocessing including SAR-optical fusion, ATPRK band fusion, and low-hallucination super-resolution—to physically enhance the input data. Second, rather than training baseline U-Net architectures from scratch, the most viable path forward is to fine-tune a Sentinel-native geospatial foundational model, such as TerraMind, using expert-curated training labels. This approach systematically bypasses spatial constraints while remaining within standard computational resource boundaries.

Table 1: Comparison of Applicable Methodologies for Road Extraction

Paper	Base Model	Data Source	Key Contribution	Primary Limitation
Zhang et al [27].	U-Net	Sentinel-1 (SAR)	Pioneered the application of deep neural networks specifically for road segmentation on SAR imagery.	Evaluation methodology is opaque; the construction process for the evaluation dataset is not outlined.
Ayala et al. [1]	U-Net & ResNet-34	Sentinel-1 (SAR), Sentinel-2 (Optical), & OpenStreetMap	Demonstrated that models 1. U-Net trained on fused SAR and optical data outperformed single-modality baselines. 2. Models can directly predict super-resolved masks, thereby improving masks quality. 3. Creation of dataset from rasterised OSM masks, filtered by human experts.	Unfiltered OSM ground truth introduces labelling noise and omission errors.
Lin et al. [14]	U-Net (United U-Net)	Sentinel-1 (SAR) & Sentinel-2 (Optical)	Fused SAR and optical modalities at the encoder level; outperformed single-modality baselines.	Lacks explicit ablation demonstrating the isolated effect of the fusion mechanism.
Sirko et al. [21]	HRNet (Teacher-Student)	Sentinel-2 paired with ultra-high res aerial	Successfully distilled labels from a high-resolution teacher model to a low-resolution student model.	Strictly requires access to parallel, corresponding high-resolution imagery for training.
Hetang et al [7].	SAM Encoder + Geometry Decoder	SpaceNet & CityScale (Ultra-high res)	Achieved state-of-the-art road graph extraction by pairing a pretrained SAM encoder with topology decoders.	Highly dependent on ultra-high-resolution datasets; applicability to 10m Sentinel data is untested.

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